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RESEARCH SUMMARY

MSc Cognitive Science student working in computational neuroimaging and the evaluation of medical AI. Research trainee at imec-Vision Lab, University of Antwerp (Prof. Jan Sijbers), on cross-cohort generalization in MRI-based Alzheimer's disease classification. My work centres on whether a model that scores well on one cohort can be trusted on another, and on the evaluation practices that decide the answer: preprocessing parity between cohorts, confounds that survive it, calibration under distribution shift, and quality-control gates that state their own coverage. Seeking a PhD in medical image analysis, neuroimaging AI or computational neuroscience.

RESEARCH EXPERIENCE

Research Trainee — imec-Vision Lab, University of Antwerp

2026 – Present

Supervisor: Prof. Jan Sijbers · 8× A100 GPU cluster

- Found and quantified a preprocessing parity defect between cohorts that had produced three separate misleading results at once — an apparent benefit from feature harmonization, an apparent advantage for a pre-trained brain foundation model, and depressed transfer. Both data versions were retained, so the before/after is a controlled contrast: zero-shot transfer rose from **0.671 ± 0.089 to 0.833 ± 0.010** and seed-to-seed variability fell 8.9-fold.
- Showed that feature harmonization removes the signal it targets without improving the objective it stands for: across three leave-one-cohort-out folds (ADNI, OASIS-3, AIBL), DeepComBat cut cross-cohort site decodability from **0.82 to 0.59** while disease AUC moved by 0.001–0.012 and calibration worsened in every arm-versus-baseline comparison.
- Ran an independent, protocol-matched test of a published brain foundation model's generalization claim (BrainIAC, Nat. Neurosci. 2026). With both encoders frozen behind the same linear probe, the foundation model **trails a from-scratch 3D CNN by 0.078 and 0.087 AUC** — the reverse of what the defective preprocessing had suggested.
- Decomposed the residual gap against per-component controls rather than a common baseline, including a size-matched control that separates population matching from training-set size. Population shift does not survive its own control (-0.024 ± 0.027); the operational definition of "Alzheimer's disease" is the only component exceeding its seed spread ($+0.031 \pm 0.012$).
- Built the evaluation to be falsifiable: five seeds per configuration, both transfer directions, subject-clustered bootstrap intervals, paired DeLong tests, and AUPRC and calibration reported alongside AUC at 10% target prevalence. Validated the choice of resampling unit by Monte-Carlo simulation rather than assertion.
- Wrote the cross-cohort quality-control battery the project depends on — intensity and registration parity, per-volume outlier detection, and gates that print what they checked and what they did not. Released as an open-source toolkit with synthetic demo data so it runs without access to clinical MRI.
- Engineering: reproducible PyTorch pipeline, DICOM→NIfTI conversion, ANTs affine registration to MNI152, HD-BET skull stripping, multi-GPU sharded preprocessing, resumable training, and figures generated from run logs so they cannot drift from the numbers they report.
- First-author manuscript in preparation, supervised by Prof. Jan Sijbers.

SOFTWARE AND RESEARCH OUTPUTS

prf-vision-benchmark — independent implementation and benchmark of the 2D-Gaussian population receptive field model

MIT licensed · archived with [DOI 10.5281/zenodo.21726011](https://doi.org/10.5281/zenodo.21726011) · github.com/parsatahmasebi7/prf-vision-benchmark

- Implemented the Dumoulin & Wandell (2008) pRF model from scratch in NumPy/SciPy and compared it against the published vistasoft solutions distributed with OpenNeuro ds003787 (NYU Retinotopy).
- Position estimates agreed at $r = 0.96$ (range 0.85–0.99) but size estimates at only $r = 0.66$ (range 0.46–0.80), each computed across 2148–3113 V1 vertices within a hemisphere and averaged over 6 hemispheres from 3 subjects. Four controls — data quantity, hemisphere, HRF specification, HRF fitting — left the asymmetry unexplained; its direction is consistent with previously reported limits on pRF size reliability.
- Built a vectorised prediction-bank fitter and verification scripts that check every documented claim resolves to the file or source it cites.

EDUCATION

MSc Cognitive Science (LM-55) — University of Messina, Italy

2024 – 2026 (expected)

Grade average 28.28 / 30 (top band) · EQF level 7

Thesis: Cross-cohort generalization in MRI-based Alzheimer's disease classification (in progress, carried out at imec-Vision Lab, University of Antwerp).

BSc Psychology — Islamic Azad University, Iran

2020 – 2024

Final grade 17.33 / 20 · EQF level 6

TECHNICAL SKILLS

Modelling / ML PyTorch, MONAI, scikit-learn, 3D CNNs, foundation models (frozen feature extraction, linear probing), transfer evaluation, encoding models, nonlinear least-squares fitting and parameter estimation

Neuroimaging structural MRI preprocessing, DICOM→NIfTI (dcm2nii), ANTs affine registration to MNI, HD-BET skull stripping, NiBabel, harmonization (ComBat, DeepComBat, adversarial gradient reversal); fMRI time-series and haemodynamic response function modelling, retinotopic mapping, surface (vertex-wise) data

Evaluation subject-clustered bootstrap intervals, paired DeLong tests, calibration by Brier score, precision-recall at low prevalence, confound and scanner decodability, subject-wise splits, leakage-free design, Monte-Carlo validation of statistical choices, agreement between independent implementations

Computing Python (NumPy, SciPy, pandas), vectorised numerical code, Git, Linux, CUDA, multi-GPU and HPC (A100 cluster), tmux-based long-running job management

LANGUAGES

English — C1 reading / B2 overall · Italian — B1/B2 · Persian — native